Ruolin Li

ruolinl@usc.edu (213) 821-4101
ruolinli.me Google Scholar
University of Southern California

WiSE Gabilan Assistant Professor
Civil and Environmental Engineering | Computer Science (courtesy)

Research Summary

I develop game-theoretic, control, and learning foundations for **societally-aware interactive autonomy**: autonomous agents that make strategic decisions while improving system-level outcomes. My work connects multi-agent systems and optimization with transportation, robotics, and emergency response, spanning mixed-autonomy traffic control, mobility markets, and resilient coordination of human–AI collectives. I direct the **Societally-aware Interactive Autonomous Systems (SIAS) Lab** at USC.

Research areas: mixed-autonomy transportation; game-theoretic control; multi-agent and agentic AI; mobility markets and infrastructure; resilient decision systems; control, optimization, and machine learning.

Selected Highlights

- Lead Principal Investigator of a four-year, approximately $2M NSF FIRE project on agentic AI for wildfire evacuation.
- Recipient of the USC WiSE Gabilan Assistant Professorship and Rising Star selections from MIT CEE and Stanford ME.
- 2026 BTR Best Paper Award; finalist for the IEEE ITSS Best Dissertation Award.
- 17 journal/conference papers and manuscripts, 20+ invited talks, and 12 doctoral, master's, and undergraduate researchers mentored.

Academic Appointments

University of Southern California Aug. 2024–present

WiSE Gabilan Assistant Professor, Civil and Environmental Engineering
Assistant Professor, Computer Science (courtesy; since Feb. 2026)

Stanford University 2023–2024

CARS Postdoctoral Fellow, Aeronautics and Astronautics; Advisor: Marco Pavone

Education

University of California, Berkeley 2018–2023

Ph.D., Mechanical Engineering (Controls); minors in Transportation Systems and Machine Learning
Dissertation: *Exploring the Potential of Autonomous Vehicles in Mixed Autonomy Transportation Systems*; Advisor: Roberto Horowitz

University of California, Berkeley 2016–2018
M.S., Mechanical Engineering (Controls); Advisor: Roberto Horowitz

Jilin University, China 2012–2016
B.S., Automotive Engineering, summa cum laude

Research Grants and Sponsored Projects

Years	Sponsor and Project	Role	Status	Amount
2026–30	**NSF FIRE Award #2623936 – EVAC-AI.** Agentic AI and multi-agent decision support for community-aware wildfire evacuation.	Lead PI	Awarded	~$2M total
2025–26	**National Center for Sustainable Transportation (NCST), U.S. DOT.** Mixed-autonomy routing with competing autonomous-vehicle companies; equilibrium modeling, computation, and policy analysis.	PI	Awarded	–

Selected Honors and Awards

Best Paper Award, 8th Bridging Transportation Researchers Conference; MAPLE paper 2026
Finalist, IEEE Intelligent Transportation Systems Society Best Dissertation Award 2024
WiSE Gabilan Assistant Professorship, University of Southern California 2024
CARS Postdoctoral Fellowship, Stanford Center for Automotive Research 2023
Rising Star in Mechanical Engineering, Stanford University 2022
Rising Star in Civil and Environmental Engineering, Massachusetts Institute of Technology 2021
Helene M. Overly Memorial Scholarship, WTS Foundation 2021
Educational Foundation and Aerospace/Test Measurement Division Scholarships, ISA 2021
Graduate Division Block Grant Award, UC Berkeley 2017, 2021–22
Outstanding Graduate, Jilin University 2016
University First Prize Scholarship, Jilin University 2013–15

Publications

* corresponding author; † student advised or co-advised by Li.

Manuscripts and Preprints

[P4] Kexin Wang†, Gavin Huang†, Zehao Wang, Jiachen Li, and **Ruolin Li***. "MAPLE: Macro–Micro Aligned Pigouvian Lane Control for Mixed-Autonomy Traffic Efficiency." Under review, 2026. **Best Paper Award, 8th BTR Conference**.

[P3] Kexin Wang†, Haohui He, and **Ruolin Li***. "When Altruism Meets Autonomy: Managing Bottleneck Congestion with Strategic Autonomous Vehicles." Under review, 2026.

[P2] Jiaxin Hou†, Kexin Wang†, **Ruolin Li***, and Jong-Shi Pang. "Traffic Equilibrium in Mixed-Autonomy Networks with Capped Customer Waiting." Under review, 2025.

[P1] **Ruolin Li**, Philip N. Brown, and Roberto Horowitz. "A Unified Toll Lane Framework for Autonomous and High-Occupancy Vehicles in Interactive Mixed Autonomy." Under review, 2024.

Peer-Reviewed Journal Articles

[J3] **Ruolin Li**, Negar Mehr, and Roberto Horowitz. "Submodularity of Optimal Sensor Placement Problems for Traffic Networks." *Transportation Research Part B: Methodological*, 171:29–43, 2023.

[J2] Stanley W. Smith, Yeojun Kim, Jacopo Guanetti, **Ruolin Li**, Roya Firoozi, Bruce Wootton, Alexander A. Kurzhanskiy, Francesco Borrelli, Roberto Horowitz, and Murat Arcak. "Improving Urban Traffic Throughput with Vehicle Platooning: Theory and Experiments." *IEEE Access*, 8:141208–141223, 2020.

[J1] Negar Mehr, **Ruolin Li**, and Roberto Horowitz. "A Game-Theoretic Macroscopic Model of Lane Choices at Traffic Diverges with Applications to Mixed-Autonomy Networks." *Transportation Research Part B: Methodological*, 144:45–59, 2020.

Peer-Reviewed Conference Papers

[C10] Haohui He, Kexin Wang†, and **Ruolin Li**. "To Stay or to Bypass: Unraveling Mainline Vehicles' Strategic Decision-Making at Highway Weaving Ramps." IEEE Intelligent Transportation Systems Conference (ITSC), 2025.

[C9] Zhilin Zhang, Y. Sun, Z. Wang, Y. Nie, X. Ma, P. Sun, and **Ruolin Li**. "Large Language Models for Mobility in Transportation Systems: A Survey on Forecasting Tasks." Transportation Research Board Annual Meeting, 2025.

[C8] Robert Dyro, Matthew Foutter, **Ruolin Li**, Luigi Di Lillo, Edward Schmerling, Xilin Zhou, and Marco Pavone. "Realistic Extreme Behavior Generation for Improved AV Testing." IEEE International Conference on Robotics and Automation (ICRA), 2025.

[C7] **Ruolin Li**and Roberto Horowitz. "Dynamic Routing and Queueing for Human-Driven and Autonomous Vehicles with Responsive Signal Control." American Control Conference (ACC), pp. 144–155, 2022.

[C6] **Ruolin Li**, Philip N. Brown, and Roberto Horowitz. "A Highway Toll Lane Framework that Unites Autonomous Vehicles and High-Occupancy Vehicles." IEEE ITSC, pp. 3482–3489, 2021.

[C5] **Ruolin Li**, Philip N. Brown, and Roberto Horowitz. "Employing Altruistic Vehicles at On-Ramps to Improve the Social Traffic Conditions." ACC, pp. 4547–4552, 2021.

[C4] **Ruolin Li**, Negar Mehr, and Roberto Horowitz. "The Impact of Autonomous Vehicles' Headway on the Social Delay of Traffic Networks." IEEE Conference on Decision and Control (CDC), pp. 268–273, 2020.

[C3] **Ruolin Li**, Jiaxi Liu, and Roberto Horowitz. "Game-Theoretic Model for Aggregate Lane Choice Behavior of Highway Mainline Vehicles at the Vicinity of On-Ramps." ACC, pp. 5376–5381, 2020.

[C2] **Ruolin Li**, Negar Mehr, and Roberto Horowitz. "An Extended Game-Theoretic Model for Aggregate Lane Choice Behavior of Vehicles at Traffic Diverges with a Bifurcating Lane." IEEE ITSC, pp. 2226–2231, 2019.

[C1] Negar Mehr, **Ruolin Li**, and Roberto Horowitz. "A Game-Theoretic Model for Aggregate Bypassing Behavior of Vehicles at Traffic Diverges." IEEE ITSC, pp. 1968–1973, 2018.

Workshop Paper, Technical Report, and Dissertation

[W1] Jiwoo Kim, Hyunwoo Ryu, Jongeun Choi, Joohwan Seo, Nikhil Potu Surya Prakash, **Ruolin Li**, and Roberto Horowitz. "Robotic Manipulation Learning with Equivariant Descriptor Fields." RSS Workshop on Symmetries in Robot Learning, 2023.

[T1] **Ruolin Li**. *SUMI: An Arterial Traffic Performance Measure Tool.* UC Berkeley, 2018.

[D1] **Ruolin Li**. *Exploring the Potential of Autonomous Vehicles in Mixed Autonomy Transportation Systems.* Ph.D. dissertation, UC Berkeley, 2023.

Invited Talks

Societally-Aware Autonomy: Games, Control, and Infrastructure for the Next Generation of Mobility Systems

– UCLA Civil and Environmental Engineering Apr. 2026

– UC Irvine Civil and Environmental Engineering Apr. 2026
– USC Industrial and Systems Engineering Feb. 2026
– USC Computer Science Oct. 2025
– Florida State University Computer Science Sept. 2025

When Altruism Meets Autonomy: Managing Weaving-Ramp Congestion with Strategic CAVs
– Bridging Transportation Researchers Conference; ICCOPT Aug. 2025

Harnessing Autonomous Vehicles for Smarter Traffic Management
– ACC Workshop on Mixed Autonomy in Transportation July 2025
– UC Riverside Electrical and Computer Engineering May 2025
– USC Information Sciences Institute Oct. 2024

A Unified Toll Lane Framework for Interactive Mixed Autonomy
– INFORMS Workshop on Societally-aware Mobility Systems Oct. 2024

The Potential of Autonomous Vehicles in Future Intelligent Transportation Systems
– USC CE/Planning and ITE; ETH Autonomy Talks; Stanford CARS; Google; UMBC; USC; Colorado Springs; Michigan; Wisconsin–Madison; MIT CEE Rising Stars 2021–24

Teaching

CE 513: Game Theory for Interactive Autonomy, USC; designer and instructor; evaluation 4.6/5 Spring 2026
CE 471: Principles of Transportation Engineering, USC; instructor Fall 2025
CE 599: Game Theory for Interactive Autonomy, USC; designer and instructor; evaluation 4.6/5 Spring 2025
ME 132: Dynamic Systems and Feedback, UC Berkeley; graduate student instructor Fall 2022
E 7: Introduction to Computer Programming for Scientists and Engineers, UC Berkeley; graduate student instructor Spring 2022
ECON C110: Game Theory in Social Sciences, UC Berkeley; evaluation 6.13/7 (department mean 5.73/7) Fall 2021

Advising and Mentoring

Ph.D. students: Kexin Wang (USC CEE, 2025–present); Jiaxin Hou (USC ISE, 2024–present; co-advised with Jong-Shi Pang).
Master's students: Haohui He (USC EE, 2024–2026); Elinor Zhang (USC Data Science, 2024–2025); Yanqi Luo (USC Urban Planning, 2024–2025); Mathias Wulfman (UC Berkeley ME, 2018–2019).
Undergraduate researchers: Brian Shi, Gavin Huang, Tyler Davis, Alina Du, Roda Hassan, and Shreya Nair (USC, 2024–present).
Selected outcome: Kexin Wang received a 2026 BTR Best Paper Award for the student-led MAPLE paper on mixed-autonomy lane control.

Professional Leadership and Service

Leadership: Vice Chair for Social Media, IFAC Technical Committee 9.2 on Social Impact of Automation (2026–present); Organizing Committee, 6th IFAC Workshop on Cyber-Physical Human Systems (2026); Session Chair, ICCOPT (2025).
NSF panels: CMMI CIS; Smart and Connected Communities; CMMI ISP; Cyber-Physical Systems; M3X; CDS&E.
Journal review: *IEEE TAC*; *IEEE RA-L*; *IEEE T-ITS*; *Transportation Science*; *IEEE TCNS*; *IEEE TCST*; *International Journal of Transportation Science and Technology*.
Conference review: CDC, ACC, ECC, ITSC, AAAI, and ICRA.

USC service: CEE Seminar Committee (2024–present); CEE Ph.D. Achievement Award Committee (2025); member of six doctoral qualifying committees across CEE and CS (Milad Pooladsanj, Yuehan Qin, Junjie Ye, Jiageng Mao, Jiawei Yang, and Li Li); thesis proposal/dissertation committees for Milad Pooladsanj and Curtis Bechtel.

Broader Impact and Outreach

Founder and director of SIAS Lab; lead development of EVAC-AI at USC, connecting multi-agent AI with wildfire evacuation and regional resilience partners; instructor for UC Berkeley SPLASH, an outreach program for students in grades 9–12 (2023).

Selected Public Scholarship and Media

Featured researcher, "Could Autonomous Vehicles Make Us More Altruistic?," USC Viterbi School of Engineering (2024); invited recorded speaker, ETH Autonomy Talks (2024); featured in USC Viterbi's New Faculty Introductions (2024). Research outreach also includes engagement with regional transportation and resilience stakeholders through EVAC-AI and mobility-systems projects.